

Chemistry (OG)

Some Basic Concepts of Chemistry

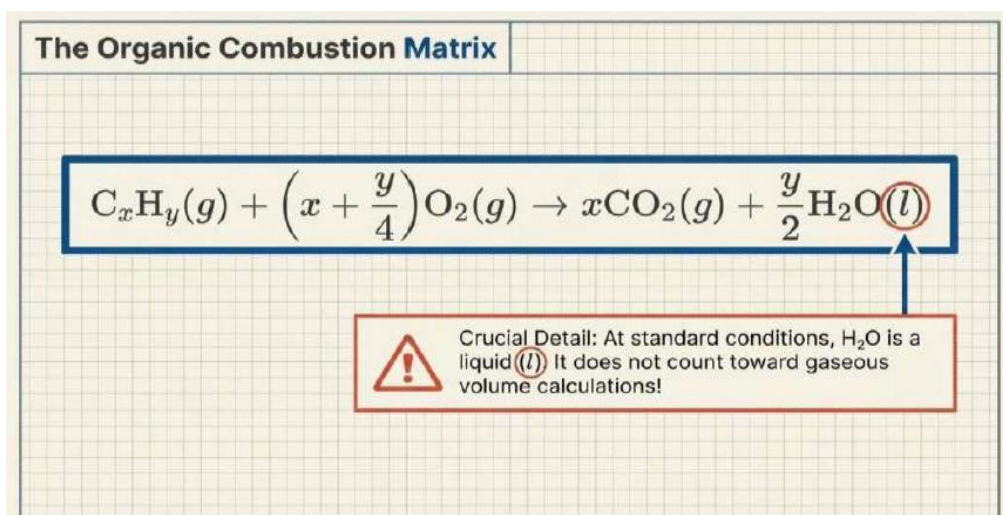
Lecture-06

Concepts Covered in the Class:

- Balancing of Combustion of Organic Compounds
- Volume Contraction
- Percentage Yield (%age Yield)
- Percentage Purity (%age Purity)

Balancing of Combustion of Organic Compounds

- Gaseous hydrocarbon combustion involves reacting a hydrocarbon with oxygen gas to produce carbon dioxide gas and liquid water.
- Balanced chemical equation: $C_xH_y(g) + \left(x + \frac{y}{4}\right) O_2(g) \rightarrow xCO_2(g) + \frac{y}{2} H_2O(l)$
- Since water is produced in the liquid state under standard conditions, its volume is considered negligible and is omitted from gaseous volume calculations.



Volume Contraction

- Volume contraction is the reduction in the total volume of gaseous species during a combustion reaction.
- Formula for 1 L of reactant gas:

□ **Hydrocarbon (C_xH_y):** $\Delta V = \left(1 + \frac{y}{4}\right) L$

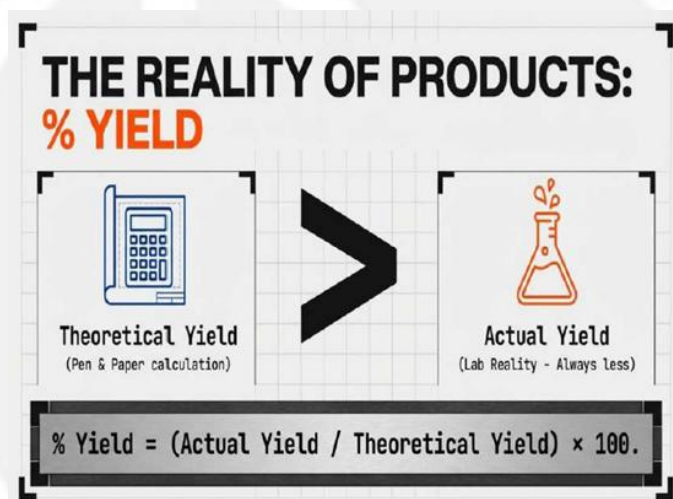
□ **Nitrogen-containing organic gas ($C_xH_yN_z$):** $\Delta V = \left(1 + \frac{y}{4} - \frac{z}{2}\right) L$

□ **Oxygen-containing organic gas ($C_xH_yO_z$):** $\Delta V = \left(1 + \frac{y}{4} - \frac{z}{2}\right) L$

- This calculation strictly accounts only for gaseous reactants and gaseous products; liquid water does not contribute to the final gaseous volume.

Percentage Yield

- Percentage yield measures the efficiency of a chemical reaction by comparing the actual quantity of product obtained to the maximum theoretical amount expected.
- Percentage Yield = $\left(\frac{\text{Actual Yield}}{\text{Theoretical Yield}} \right) \times 100$
- In practice, the actual yield is always less than the theoretical yield due to incomplete reactions, side reactions, or mechanical losses during product isolation.



Percentage Purity

- Percentage purity defines the ratio of the mass of the active, pure chemical reactant to the total mass of the impure sample.
- Percentage Purity = $\left(\frac{\text{Mass of pure substance}}{\text{Total mass of sample}} \right) \times 100$
- Stoichiometric calculations must only be performed using the mass of the pure substance, as inert impurities do not participate in the chemical reaction.



Concept 1: Percentage Purity (The Reality of Reactants)

Reactants are rarely 100% pure in the real world. Only the pure fraction participates in the chemical reaction.

Bhoosa (Impurity) = 98g

100g

Dimaag (Pure) = 2g

The Golden Rule
All calculations must be done according to the mass of the pure substance.

% Purity = $\frac{\text{Mass of Pure Substance}}{\text{Total Mass of Sample}} \times 100$

So if it's only 2% pure, I only use 2g for the calculation?
Exactly! The 98g is just filler, ignore it!